

Prokaryotic Promoter Structure

Reference: Molecular Biology, 5th Edition, Robert F. Weaver

1) Prokaryotic Promoter Structure

❖ Core Promoter Elements:

○ Pribnow box/ -10 box:

- Region Common To *E. Coli* And Phage Promoters
- A Sequence Of 6 Or 7 Bp Centered Approximately 10 Bp Upstream Of The Start Of Transcription

○ -35 Box:

1. Noticed by Mark Ptashne and colleagues
2. short sequence centered approximately 35 bp upstream of the transcription start site

❖ Consensus Sequence:

- Thousands of promoters have now been examined and a typical, or **consensus sequence** for each of these boxes has emerged.
- These so-called consensus sequences represent probabilities.
- The **capital letters** in Figure below denote bases that have a high probability of being found in the given position in **more than 50% of promoters examined**.
- The **lowercase letters** correspond to bases that are usually found in the given position in **50% or fewer of promoters examined**.
- -10 or -35 boxes that match the consensus sequences perfectly are rarely found
- However, when such perfect matches are found, they tend to occur in very strong promoters that initiate transcription unusually actively.
- **Down Mutations:** destroy matches with the consensus sequences, make promoter weaker.
- **Up Mutations:** make the promoter sequences more like the consensus sequences, usually make the promoters stronger.
- spacing between promoter elements is also important
- deletions or insertions that move the -10 and -35 boxes unnaturally close together or far apart are deleterious

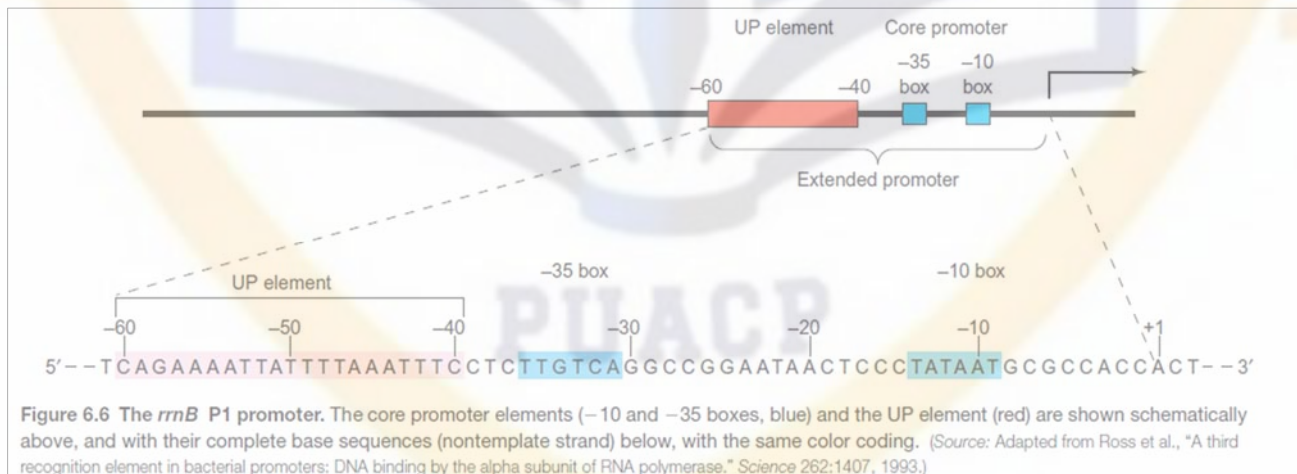


❖ UP Element:

- In some very strong promoters i.e. in extraordinarily powerful promoters of seven *E. coli* genes (***rrn* genes**) that encode rRNAs.
- Under rapid growth conditions, when rRNAs are required in abundance, these seven genes by themselves account for the majority of the transcription occurring in the cell.

❖ Structure of *rrnB* P1 promoter:

- Upstream of the core promoter (blue), there is an UP element (red) between positions -40 and -60.
- UP element is a true promoter element because it stimulates transcription of the *rrnB* P1 gene by a factor of 30 in the presence of RNA polymerase alone.
- Because it is recognized by the polymerase itself, we conclude that it is a promoter element.
- **Fis sites:** associated with this promoter, between positions -60 and -150, binding sites for the transcription-activator protein, called **enhancers** being members of another class of transcription-activating DNA elements that do not bind to RNA polymerase itself.



2) Regulation of *E. coli rrn* promoters:

- regulated by a pair of small molecules:
 1. the initiating NTP (the iNTP)
 2. guanosine 59-diphosphate 39-diphosphate (ppGpp), an **alarmone**.

- An **abundance of iNTP** indicates that the concentration of nucleotides is high, and therefore it is appropriate to synthesize plenty of rRNA. Accordingly, iNTP stabilizes the open promoter complex, **stimulating transcription**.
- When **cells are starved for amino acids**, protein synthesis cannot occur readily and the **need for ribosomes (and rRNA) decreases**. Ribosomes sense the lack of amino acids when uncharged tRNAs bind to the ribosomal site where aminoacyl-tRNAs would normally bind. Under these conditions, a ribosome-associated protein called **RelA receives the “alarm” and produces the “alarmone” ppGpp**, which **destabilizes open promoter complexes** whose lifetimes are normally short, thus **inhibiting transcription**.
- **Role of DskA:**
 - ✓ plays an important role
 - ✓ binds to RNA polymerase and reduces the lifetimes of the *rrn* open promoters to a level at which they are responsive to changes in iNTP and ppGpp concentrations.
 - ✓ Thus, DskA is **required for the regulation of *rrn* transcription by these two small molecules**.
 - ✓ Indeed, *rrn* transcription is insensitive to iNTP and ppGpp in mutants lacking DskA.